

Paper I - Basic Sciences in Urology (MCh Urology, Summer 2015)

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I-1. Describe in details basic patho-physiology of urine production and transport

Type: LAQ (25 marks)

Introduction

- Formation = filtration → reabsorption → secretion → concentration/dilution; transport = active pelvi-ureteric peristalsis → compliant bladder.

A. Renal blood flow (RBF)

- RBF ≈ **20-25% of cardiac output (CO)** (~1-1.2 L/min); renal plasma flow (RPF) ~600 mL/min; cortex ~90%, medulla <10% (protects gradient; hypoxia-prone).
- **Autoregulation** keeps RBF and GFR (glomerular filtration rate) constant over mean arterial pressure **80-180 mmHg**:
 - Myogenic – afferent stretch → contraction.
 - **Tubuloglomerular feedback (TGF)** – ↑ NaCl at macula densa → adenosine → afferent constriction (↓ renin).
- Angiotensin II (Ang II) constricts efferent > afferent; non-steroidal anti-inflammatory drugs (NSAIDs) remove prostaglandin protection in low flow.

B. Glomerular filtration

- GFR ~**125 mL/min (~180 L/day)**; filtration fraction (FF = GFR/RPF) ≈ **20%**.
- **Starling:** $GFR = K_f \times [(P_{GC} - P_{BS}) - (\pi_{GC} - \pi_{BS})]$. Glomerular capillary hydrostatic pressure = main determinant; Bowman's space pressure ↑ in **obstruction**; K_f = surface area (mesangial contraction) + permeability.
- Barrier: fenestrated endothelium → basement membrane (negative charge) → podocyte slit diaphragm; restricts size (>~60-70 kDa) and charge (albumin excluded).
- Measurement: inulin (gold standard), creatinine clearance, DTPA (diethylenetriamine pentaacetic acid) renography.

C. Tubular handling

Segment	Key transporters	Handling	Clinical link
Proximal convoluted tubule (PCT)	Na ⁺ /H ⁺ exchanger, SGLT2 (sodium-glucose cotransporter 2), carbonic anhydrase	~ 65% Na ⁺ /water (iso-osmotic), ~100% glucose, ~80-90% HCO ₃ ⁻ ; secretes drugs, creatinine	Acetazolamide, SGLT2 inhibitors; Fanconi
Thick ascending limb (TAL)	NKCC2 (Na ⁺ -K ⁺ -2Cl ⁻ cotransporter), ROMK (renal outer medullary K ⁺ channel)	~ 25% NaCl; water-impermeable "diluting segment" ; builds medullary gradient	Loop diuretics; Bartter
Distal convoluted tubule (DCT)	NCC (Na ⁺ -Cl ⁻ cotransporter)	~5-7% Na ⁺ ; Ca ²⁺ reabsorption (parathyroid hormone, PTH)	Thiazides ↓ urinary Ca (hypercalciuric stones); Gitelman
Collecting duct (CD) - principal cells	ENaC (epithelial Na ⁺ channel), aquaporin-2 (AQP2)	Na ⁺ reabsorption/K ⁺ secretion (aldosterone); water (antidiuretic hormone, ADH)	Amiloride, spironolactone; Liddle
Intercalated cells	A: H ⁺ -ATPase; B: pendrin	Acid-base	Distal renal tubular acidosis (RTA) → calcium phosphate stones, nephrocalcinosis

D. Concentration and dilution

- **Countercurrent multiplication:** TAL pumps NaCl without water → single effect (~200 mOsm) multiplied → papillary interstitium ~**1200 mOsm/kg**.
- **Urea recycling** (ADH-stimulated urea transporters, inner medullary CD) → ~half of inner-medullary osmolality.
- **Vasa recta** (countercurrent exchange) preserve gradient.
- **ADH:** ↑ osmolality (~1-2% change) or ↓ volume → **V2 receptor** → Gs → cAMP (cyclic adenosine monophosphate) → protein kinase A → **AQP2 insertion** into apical membrane → water reabsorption.
 - Endothelial V2 → von Willebrand factor/factor VIII → **desmopressin** (nocturia, enuresis, bleeding).
 - Urine osmolality ~**50-1200 mOsm/kg**.
 - Nephrogenic diabetes insipidus (DI): lithium, hypercalcaemia, hypokalaemia, **post-obstructive** (↓ AQP2).

E. Upper urinary tract transport

- **Pacemaker:** proximal pelvis/pelvicalyceal border; **atypical smooth muscle cells (SMCs)** = primary pacemakers; **interstitial cells of Cajal (ICC)-like cells** (c-Kit⁺) = secondary pacemakers/conductors (↓ in ureteropelvic junction (UPJ) obstruction). Activity **myogenic** – persists after denervation/transplant; nerves only modulate.
- **Coupling:** L-type Ca²⁺ action potential; gap junctions (syncytium) ~2-6 cm/s; Ca²⁺-calmodulin → myosin light-chain kinase → contraction.
- **Bolus:** ureter contracts behind bolus, walls **coapt**. Peristalsis **2-6/min**; resting pressure ~0-5 cmH₂O; contraction ~20-80 cmH₂O. Low flow → "pyeloureteric block".
- **Flow:** ↑ flow → ↑ frequency, then ↑ bolus volume → very high flow: boluses **coalesce** → fluid column → **dilatation without obstruction** (diuresis; diuretic renogram).
- **Ureterovesical junction (UVJ):** oblique tunnel, **length:diameter ≈ 5:1** (Paquin) prevents reflux; sustained bladder pressure >~40 cmH₂O (neurogenic bladder) → efflux fails → secondary hydronephrosis.

Factor	Effect
Infection	Gram-negative endotoxin inhibits peristalsis → dilatation
Pregnancy	Dilatation R > L; mechanical ± progesterone
Drugs	Relax: α1-blockers (tamsulosin – medical expulsive therapy), Ca-channel blockers, β-agonists; NSAIDs ↓ pelvic pressure (renal colic)

F. Obstruction

Unilateral ureteral obstruction (UUO) - triphasic:

Phase 1 (0–1.5 h):	RBF ↑, pressure ↑	(afferent dilatation: prostaglandin E2, nitric oxide)
Phase 2 (1.5–5 h):	RBF ↓, pressure ↑	(efferent constriction)
Phase 3 (>5 h):	RBF ↓, pressure ↓	(afferent constriction: Ang II, thromboxane, endothelin)

- GFR ↓ (↓RBF + ↑tubular pressure); contralateral kidney compensates.
- **Bilateral/solitary:** pressure stays high longer; natriuretic substances (atrial natriuretic peptide, urea) accumulate → **post-obstructive diuresis**.
- Tubular: ↓ concentration (↓AQP2), ↓ Na reabsorption, distal RTA, ↓ K⁺ secretion.
- Chronic: Ang II + transforming growth factor-β1 → fibrosis; recovery depends on duration, degree, infection.
- **Pyelorenal backflow** = pop-off; forniceal rupture → urinoma.

Diagram to draw: Nephron with segmental transporters/% Na⁺ + ADH-AQP2 inset; UUO triphasic RBF/pressure graph.

Recent advances

- SGLT2 inhibitors restore TGF (afferent constriction) → nephroprotection.
- Tolvaptan (V2 antagonist) – aquaresis in autosomal dominant polycystic kidney disease.

I-2. Describe female pelvic anatomy in relation to prolapse and SUI

Type: LAQ (25 marks)

Introduction

- Support = **levator ani + endopelvic fascia/ligaments + bony pelvis**. Failure → **pelvic organ prolapse (POP)** (anterior/apical/posterior descent) and **stress urinary incontinence (SUI)** (effort leakage when urethral support/closure fails).

A. Bony pelvis

- Standing: organs rest on **levator plate**, not hiatus. Landmarks: **ischial spine**, **sacrospinous** and **Cooper's ligaments**, sacral promontory.

B. Pelvic floor (pelvic diaphragm)

Muscle	Attachment	Function
Pubococcygeus/pubovisceral	Pubis → vagina, perineal body, anus	Closes hiatus, lifts urethra/vagina
Puborectalis	Sling behind anorectal junction	Anorectal angle; closes hiatus
Iliococcygeus	Arcus tendineus levator ani (ATLA) → coccyx	Horizontal levator plate

- Levator avulsion** (vaginal delivery; 3D/4D translabial ultrasound or MRI (magnetic resonance imaging)) and ↑ hiatal area → POP and recurrence.

C. Endopelvic fascia and ligaments

- Cardinal & uterosacral** – apical suspension.
- Pubocervical fascia** – anterior wall, attached laterally to **arcus tendineus fasciae pelvis (ATFP, white line)** (pubis → ischial spine).
- Rectovaginal fascia (Denonvilliers)** – posterior; perineal body to uterosacral complex.
- Pubourethral ligaments** – mid-urethra to pubis.

D. DeLancey's levels of support (1992)

Level	Region	Structures	Defect →
I - suspension	Cervix/upper vagina	Cardinal-uterosacral	Uterine/vault prolapse, enterocele
II - attachment	Mid vagina	ATFP (anterior), ATLA/rectovaginal fascia (posterior)	Cystocele (paravaginal), rectocele
III - fusion	Distal 2–3 cm	Perineal membrane, perineal body, levator	Urethral hypermobility, perineal descent, low rectocele

E. Continence mechanisms

- Hammock hypothesis (DeLancey 1994)**: urethra rests on anterior vaginal wall + endopelvic fascia ; ↑ abdominal pressure compresses urethra against it. Loss → SUI. Replaced Enhörning's pressure-transmission theory.
- Integral theory (Petros & Ulmsten 1990)**: stress and urge both from laxity of vagina/ligaments; **mid-urethral pubourethral ligament** key → basis of **tension-free vaginal tape (TVT, 1996)**.

- **Intrinsic closure:** mucosal coaptation (oestrogen-dependent), smooth muscle, rhabdosphincter.
- **Intrinsic sphincter deficiency (ISD):** Valsalva leak point pressure <60 cmH₂O / maximal urethral closure pressure <20 cmH₂O → poorer colposuspension results; prefer sling/bulking.
- SUI = **hypermobility ± ISD** – a continuum.

F. Compartment defects

Compartment	Defect	Basis
Anterior	Central cystocele	Midline pubocervical attenuation – rugae lost
	Paravaginal cystocele	Detachment from ATRP – rugae preserved , lateral sulci lost
Apical	Uterine/vault prolapse, enterocele	Level I failure; enterocele post-hysterectomy/colposuspension
Posterior	Rectocele, perineal descent	Rectovaginal fascia defect, perineal body disruption

- **Apex underpins anterior** – anterior repair fails if apex ignored.

G. Innervation

- **Pudendal (S2-4)** – sphincters; delivery stretch injury. **Levator ani nerve (S3-5)** – pelvic surface, separate.
- Autonomic: pelvic splanchnic (S2-4), hypogastric (T10-L2) – injured in radical surgery.

H. Risk factors (Bump & Norton)

- **Predisposing:** genetics, race, connective tissue disorders (Ehlers-Danlos, Marfan).
- **Inciting: vaginal childbirth** (strongest; forceps, macrosomia, levator avulsion), hysterectomy.
- **Promoting:** obesity, chronic cough, constipation.
- **Decompensating:** ageing, **menopause**, neuropathy.

I. POP-Q (Pelvic Organ Prolapse Quantification; International Continence Society, Bump 1996)

- Reference **hymen = 0**; above negative, below positive (cm). Points: Aa, Ba, C, D, Ap, Bp, gh (genital hiatus), pb (perineal body), tvl (total vaginal length).

Stage	Definition
0	No prolapse
I	Most distal point >1 cm above hymen
II	Within ±1 cm of hymen
III	>+1 cm below hymen but < (tvl – 2)
IV	Complete eversion (≥ tvl – 2)

Clinical assessment

- Valsalva standing; Sims speculum; **cough stress test** (after reducing prolapse – occult SUI); **Q-tip** >30° = hypermobility; urodynamics if ISD suspected.

J. Anatomy guides surgery

Defect	Operation	Principle
SUI – hypermobility	Mid-urethral sling (MUS) (retropubic TVT/transobturator)	Recreates pubourethral support
SUI	Burch colposuspension	Periurethral fascia → Cooper's ligament
SUI – ISD	Autologous pubovaginal sling, bulking, artificial urinary sphincter	Compression/coaptation
Paravaginal cystocele	Paravaginal repair	Reattach to ATRP
Central cystocele	Anterior colporrhaphy	Midline plication
Apical	Sacrocolpopexy (gold-standard durability)	Mesh → anterior longitudinal ligament over S1 (beware middle sacral vessels, left common iliac vein)
Apical, vaginal	Sacrospinous fixation (~1.5–2 cm medial to ischial spine – avoid pudendal vessels/nerve); uterosacral suspension (ureteric risk – cystoscopy)	Level I
Rectocele	Posterior colporrhaphy, perineorrhaphy	Level II/III

K. Mesh

- **US Food and Drug Administration (FDA), April 2019:** stop sale of **transvaginal mesh for anterior POP**; **not** MUS or sacrocolpopexy mesh.
- Complications: exposure, pain, dyspareunia.

Diagram to draw: DeLancey levels I–III; hammock cross-section.

Recent advances

- 3D/4D translabial ultrasound and dynamic MRI – levator avulsion predicts recurrence.
 - Single-incision mini-slings comparable short-term to MUS (SIMS RCT (randomised controlled trial), 2022).
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I-3a. Newer generation CT scan

Type: SAQ (10 marks)

Evolution

Generation	Feature
Conventional (Hounsfield, 1971-72)	Step-and-shoot axial
Helical CT (computed tomography) (~1989)	Slip-ring, continuous table → volumetric data
Multidetector CT (MDCT) (4-slice 1998 → 320-row)	Sub-mm isotropic voxels, 3D reconstruction, one breath-hold
Dual-source CT (2005)	Two tube-detector pairs: speed, dual-energy
Photon-counting CT (PCCT)	FDA (US Food and Drug Administration) clearance 2021

Dual-energy CT (DECT)

- Two energy spectra (e.g. 80/140 kilovolt peak, kVp) → material decomposition:
 - Uric acid vs non-uric-acid stones** → oral chemolysis candidates.
 - Virtual non-contrast** images → drop unenhanced phase (↓ dose).
 - Iodine maps** – hyperdense cyst vs solid mass.
 - ↓ Metal artefact (virtual monoenergetic).

Low-dose non-contrast CT (NCCT) KUB (kidney-ureter-bladder)

- Standard ~**8-10 mSv**; low-dose <**3-4 mSv**; ultra-low <**1 mSv**.
- Stone sensitivity/specificity ~95% (lower <3 mm, body mass index >30). European Association of Urology (EAU): **low-dose NCCT** for acute flank pain (after ultrasound) and follow-up.

CT urography (CTU)

- For visible haematuria/upper tract urothelial carcinoma.
- Triple-phase:** unenhanced → nephrographic (~90-100 s) → excretory (~10-15 min); ~15-25 mSv.
- Split-bolus:** 2 contrast aliquots, one combined acquisition → ↓ dose.

Dose reduction

- Tube current modulation, low kVp; **iterative** and **deep-learning reconstruction**.

PCCT

- Detector counts individual photons and sorts energy → spectral data, **no electronic noise**, ↑ resolution, ↓ dose, better stone composition.

Other

- 4D/perfusion CT; CT angiography (donor nephrectomy, 3D models for partial nephrectomy/PCNL (percutaneous nephrolithotomy) planning); radiomics/AI (artificial intelligence) stone volumetry.

Uses

- Stones (Hounsfield units >1000 = hard stone; skin-to-stone distance >10 cm → shock wave lithotripsy failure), Bosniak 2019, staging, trauma (delayed phase), emphysematous pyelonephritis.

Diagram to draw: Timeline axial → helical → MDCT → dual-source → PCCT.

I-3b. History of Laparoscopy

Type: SAQ (10 marks)

Timeline

Year	Pioneer	Milestone
1806	Bozzini	Lichtleiter – birth of endoscopy
1853	Desormeaux	"Father of endoscopy"
1877–79	Nitze	Cystoscope with distal illumination, lens system
1901	Kelling	Insufflated abdomen of a dog
1910	Jacobaeus	First human series; coined "laparoscopy"
1911	Bernheim	First in USA ("organoscopy")
1924	Zollikofer	CO ₂ insufflation
1938	Veress	Spring-loaded needle
1950s	Hopkins	Rod-lens (with Storz cold light)
1970s–80s	Semm	Automatic CO ₂ insufflator, endoloops, pelvi-trainer; first lap appendectomy 1980
1970s	Hasson	Open access
1976	Cortesi	Impalpable testis – first urological use
1985	Mühe	First lap cholecystectomy
1980s	Video chip camera	Whole team sees image; two-handed surgery
1987	Mouret	Video-laparoscopic cholecystectomy
1990	Clayman	First lap nephrectomy
1992	Gagner	Lap adrenalectomy
1992	Schuessler	First lap radical prostatectomy (reported 1997, 9 patients)
1993	Schuessler; Kavoussi & Peters	Dismembered pyeloplasty
1995	Ratner	Live-donor nephrectomy
1998–99	Guillonneau & Vallancien	Standardised lap radical prostatectomy
2000	da Vinci	FDA (US Food and Drug Administration) clearance
2000	Binder	First robot-assisted radical prostatectomy (RARP)
Early 2000s	Menon	Vattikuti Institute Prostatectomy (VIP) – standardised RARP
2004	Kalloo	NOTES (natural orifice transluminal endoscopic surgery) concept
2007	Raman, Rane	Single-port nephrectomy
2018	da Vinci SP	Single-port robot FDA clearance (urology)

Phases

Optics → Diagnostic → Therapeutic (Semm) → Video (lap chole)
→ Urological → Robotic → Single-site/NOTES/telesurgery

Indian contribution

- Indigenous robot **SSI Mantra**; telesurgery demonstrations.

Recent advances

- Competitor robots (Versius, Hugo RAS, Hinotori); ultra-long-distance telesurgery; haptic feedback (da Vinci 5).

Diagram to draw: Timeline 1806 → 1901 → 1910 → 1938 → 1987 → 1990 → 2000.

I-3c. Renin-angiotensin system

Type: SAQ (10 marks)

Definition

- Hormonal cascade regulating **blood pressure, extracellular volume, Na^+/K^+ and GFR (glomerular filtration rate)**; systemic + intrarenal.

Cascade

- ACE = angiotensin-converting enzyme; AT1/AT2 = angiotensin II type 1/2 receptors; JG = juxtaglomerular.

Angiotensinogen (liver)
| Renin (JG cells) – rate-limiting
Angiotensin I
| ACE (lung) – also degrades bradykinin
Angiotensin II → AT1 / AT2 receptors
| ACE2 → Ang(1-7) → Mas receptor (vasodilatation, anti-fibrotic)

Juxtaglomerular apparatus

- **JG cells** (afferent arteriole) – secrete renin; baroreceptors.
- **Macula densa** – senses tubular NaCl.
- Extraglomerular mesangial (Lacis) cells.

Renin release

Stimulate	Inhibit
↓ Perfusion pressure	↑ Perfusion pressure
↓ NaCl at macula densa	↑ NaCl at macula densa
Sympathetic $\beta 1$	Ang II feedback, β -blockers
Prostaglandins, loop diuretics, hypovolaemia	ANP (atrial natriuretic peptide), NSAIDs (non-steroidal anti-inflammatory drugs)

Actions

- **AT1:** vasoconstriction; **aldosterone** → Na^+ retention, K^+/H^+ loss; ADH (antidiuretic hormone), thirst; hypertrophy; **fibrosis**.
- **AT2:** vasodilatation, antiproliferative; fetal urinary tract development (knockout → CAKUT (congenital anomalies of kidney and urinary tract)).
- **Ang II constricts efferent > afferent** → maintains GFR in low perfusion → **ACE inhibitors/angiotensin receptor blockers (ARBs) cause acute GFR fall** in bilateral renal artery stenosis (RAS)/solitary kidney, hypovolaemia, NSAIDs.

Urological relevance

- **Renovascular hypertension (HTN)** (Goldblatt 1934): atherosclerotic vs fibromuscular dysplasia; **renal vein renin ratio >1.5** predicts response to revascularisation/nephrectomy.
- **Page kidney** – perirenal compression (subcapsular haematoma) → renin-mediated HTN; drainage/nephrectomy.
- **Reninoma (JG cell tumour):** young, severe HTN, **hypokalaemia**, secondary hyperaldosteronism; **partial nephrectomy curative**.
- **Obstruction:** Ang II mediates late afferent constriction and fibrosis.

- **Drugs:** ACE inhibitors (cough, angio-oedema), ARBs, aliskiren (direct renin inhibitor); contraindicated in pregnancy.

Diagram to draw: JG apparatus + cascade flowchart.

Recent advances

- Finerenone; renal denervation (FDA (US Food and Drug Administration) 2023).
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I-3d. Patho-physiology of Brain stem death

Type: SAQ (10 marks)

Definition

- **Irreversible loss of capacity for consciousness + capacity to breathe;** legally death.
Main source of donation after brainstem death (DBD) kidneys.

Pathophysiology

- ICP/CPP = intracranial/cerebral perfusion pressure.

Brain injury → oedema → ↑ICP → ↓CPP → herniation ("coning")
→ Pons: CUSHING REFLEX (hypertension, bradycardia)
→ Medulla: CATECHOLAMINE STORM (hypertension, arrhythmia, myocardial necrosis)
→ Loss of sympathetic tone → HYPOTENSION

System	Change
Posterior pituitary	Diabetes insipidus (DI) - ↓ADH (antidiuretic hormone), hypernatraemia
Anterior pituitary	↓T3 (triiodothyronine), adrenal insufficiency
Temperature	Hypothermia
Haematology	DIC (disseminated intravascular coagulation) (brain thromboplastin)
Immune	Inflammatory response → ↑delayed graft function and rejection in DBD

Diagnosis

- **Preconditions:** apnoeic coma on ventilator, known irreversible cause. **Exclude:** sedatives, neuromuscular blockers, **hypothermia (<35 °C)**, metabolic/endocrine causes, shock.
- **Absent brainstem reflexes:** pupillary (II, III), corneal (V, VII), caloric (~50 mL ice water; VIII, III, VI), doll's eye, motor response to central pain (V, VII), gag/cough (IX, X).
- **Apnoea test:** pre-oxygenate 100% O₂, disconnect; no breathing at **PaCO₂ (arterial CO₂ tension) ≥60 mmHg** (pH <7.30). Abort if unstable.
- Ancillary tests (EEG (electroencephalography), angiography) – not mandatory in India/UK.

Indian law

- **Transplantation of Human Organs Act (THOA) 1994;** amended 2011; Rules 2014.
- **4-member board:** hospital in-charge, independent nominated doctor, neurologist/neurosurgeon, treating doctor; no transplant team member.
- **Two tests ≥6 h apart;** mandatory request.

Donor management

- **Rule of 100s:** systolic blood pressure >100 mmHg, urine >100 mL/h, PaO₂ (arterial O₂ tension) >100 mmHg, haemoglobin >100 g/L.
- **Vasopressin** preferred (treats DI); desmopressin; hormone resuscitation.
- Donor dopamine ↓dialysis (Schnuelle 2009); mild donor hypothermia ↓delayed graft function (Niemann 2015).

Diagram to draw: Herniation sequence → storm → DI/hypothermia/DIC.

I-3e. Male reproductive immunology

Type: SAQ (10 marks)

Introduction

- Sperm appear at puberty, after tolerance → germ cell antigens "foreign"; testis **immune-privileged**.

Immune privilege

- Blood-testis barrier (BTB): Sertoli-Sertoli tight junctions** (claudin-11, occludin); formed at **puberty**; **basal** (spermatogonia) vs **adluminal** (spermatocytes, spermatids) compartments. Weak points: rete testis, efferent ducts/epididymis.
- Active immunoregulation:** M2 macrophages (interleukin-10), **regulatory T cells (Tregs)**, Sertoli TGF- β (transforming growth factor- β), immunosuppressive **androgens**; Fas ligand (FasL) role **debated**; seminal plasma immunosuppression.

Antisperm antibodies (ASA)

- Causes: vasectomy** (commonest), obstruction, testicular trauma/torsion, biopsy, infection, cryptorchidism.
- IgG (serum + semen) and IgA (local, more significant) – immunoglobulin G/A.
- Effects:** agglutination, ↓ motility and **cervical mucus penetration**, ↓ acrosome reaction, zona binding failure; head-directed antibodies more significant.

Test	Principle
Mixed antiglobulin reaction (MAR)	Fresh semen + IgG/IgA-coated beads → mixed agglutinates
Immunobead test (IBT)	Washed sperm + anti-IgG/IgA/IgM beads; % and location
Indirect	ASA in serum/seminal plasma/cervical mucus (donor sperm)

- World Health Organization (WHO) 2010: $\geq 50\%$ motile sperm with bound particles** = significant.
- Indications: isolated asthenozoospermia, agglutination on semen analysis, unexplained infertility, post-trauma/reversal.

Treatment	Comment
Corticosteroids	Inconsistent; avascular necrosis – not recommended
Washing + intrauterine insemination	Bypasses mucus; modest
ICSI (intracytoplasmic sperm injection)	Preferred – fertilisation unaffected

Inflammation

- Leukocytospermia: $>1 \times 10^6$ peroxidase-positive leukocytes/mL** → reactive oxygen species → DNA (deoxyribonucleic acid) fragmentation; culture, antibiotics if infected.
- Orchitis:** mumps (post-pubertal; atrophy); bacterial – *Chlamydia/gonorrhoea* (<35 y), *E. coli* (older); granulomatous.
- Autoimmune orchitis:** primary (ASA + azoospermia) vs secondary (vasculitis).

Diagram to draw: Seminiferous tubule – Sertoli tight junctions dividing basal/adluminal compartments.

Recent advances

- Sperm DNA fragmentation testing; checkpoint inhibitor orchitis.

I-3f. Principles of antimicrobial therapy in UTI

Type: SAQ (10 marks)

Classification (EAU (European Association of Urology) 2024)

- Urinary tract infection (UTI) types:
- **Uncomplicated** – non-pregnant women, no abnormality/comorbidity.
- **Complicated** – men, pregnancy, abnormality, catheter, immunosuppression.
- **Recurrent** – ≥ 2 in 6 months or ≥ 3 in 12 months.

Principles

1. Confirm, culture (except simple cystitis in women).
2. **Pharmacokinetics/pharmacodynamics: time-dependent** (time above minimum inhibitory concentration, $T > MIC$) – β -lactams; **concentration-dependent** (peak concentration, C_{max} ; area under the curve, AUC) – **aminoglycosides, fluoroquinolones (FQs)** (high once-daily).
3. Nitrofurantoin/fosfomycin – **low tissue levels** → **cystitis only**. Avoid nitrofurantoin at $eGFR$ (estimated glomerular filtration rate) < 30 mL/min.
4. Local resistance: empirical only if **$< 20\%$** (cystitis); FQ for pyelonephritis only if **$< 10\%$** .
5. Avoid FQ/cephalosporins in cystitis (collateral damage – ESBL (extended-spectrum β -lactamase), *C. difficile*).
6. Shortest duration, de-escalate, IV (intravenous) → oral, **source control**.
7. Pregnancy: avoid FQ, tetracyclines.

Empirical choices

Scenario	First-line	Duration
Cystitis (women)	Fosfomycin 3 g; nitrofurantoin; pivmecillinam	Single; 5 d ; 3-5 d
Alternative (resistance $< 20\%$)	Trimethoprim-sulfamethoxazole (TMP-SMX)	3 d
Cystitis in men	TMP-SMX/FQ	7 d
Pyelonephritis (oral)	Ciprofloxacin; levofloxacin 750 mg	7 d; 5 d
Pyelonephritis (IV)	Ceftriaxone, gentamicin, FQ	Oral when afebrile
Complicated	Aminoglycoside combinations, 3rd-gen cephalosporin	7-14 d
Catheter-associated	Change catheter if > 2 weeks	7 d
Urosepsis	IV broad-spectrum within 1 h + source control	

Asymptomatic bacteriuria (ASB)

- Treat **only in pregnancy and before mucosa-breaching procedures**.

Prophylaxis

- Recurrent: vaginal oestrogen, OM-89, **methenamine hippurate** (ALTAR, 2022 – non-inferior to antibiotics), continuous low-dose or post-coital.
- Peri-operative: single dose/ ≤ 24 h.

Diagram to draw: Pharmacokinetic/pharmacodynamic curve (C_{max} , AUC, $T > MIC$).

Recent advances

- Pivmecillinam FDA (US Food and Drug Administration) 2024; gepotidacin 2025.

Rapid Revision Sheet - Paper I

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

I-1. Describe in details basic patho-physiology of urine production and transport (LAQ)

- RBF 20–25% CO; GFR 125 mL/min; FF 20%; autoregulation 80–180 mmHg (myogenic + TGF).
- PCT 65%, TAL 25% (diluting), DCT 5–7%; CD fine-tunes (aldosterone; ADH → V2 → cAMP → AQP2).
- Pacemaker = atypical SMCs at pelvicalyceal border; peristalsis myogenic, 2–6/min; high flow → dilatation without obstruction.
- UUU triphasic: $\uparrow/\uparrow$, $\downarrow/\uparrow$, $\downarrow/\downarrow$ (RBF/pressure).

I-2. Describe female pelvic anatomy in relation to prolapse and SUI (LAQ)

- DeLancey: I suspension (cardinal-uterosacral), II attachment (ATFP), III fusion (perineal membrane/body).
- Hammock (DeLancey 1994) vs Integral theory (Petros & Ulmsten 1990 → TVT 1996, mid-urethra).
- Central (rugae lost) vs paravaginal (rugae preserved) cystocele; POP-Q hymen = 0, Stage II ± 1 cm, Stage IV $\geq$ tvl-2.
- FDA 2019 – transvaginal mesh for anterior POP withdrawn; MUS and sacrocolpopexy mesh unaffected.

I-3a. Newer generation CT scan (SAQ)

- Helical ~1989; MDCT 1998; dual-source 2005; PCCT FDA 2021.
- DECT: uric acid vs non-uric acid, virtual non-contrast, iodine maps.
- Low-dose KUB <3 – 4 mSv, ultra-low <1 mSv – EAU preferred NCCT.
- Split-bolus CTU reduces phases and dose.

I-3b. History of Laparoscopy (SAQ)

- Bozzini 1806 Lichtleiter; Kelling 1901 (dog); Jacobaeus 1910 (human).
- Veress 1938; Hopkins rod-lens; Semm – father of modern operative laparoscopy.
- Urology: Cortesi 1976; Clayman 1990; Schuessler 1992/1997; pyeloplasty 1993; Ratner 1995.
- da Vinci FDA 2000; Binder 2000 first RARP; Menon – VIP.

I-3c. Renin-angiotensin system (SAQ)

- Renin = rate-limiting; JG cells; stimuli $\downarrow$ pressure, $\downarrow$ macula densa NaCl, $\beta 1$.
- Ang II efferent constriction maintains GFR → ACE inhibitor danger in bilateral RAS.
- AT1 = pressor/fibrotic; AT2/ACE2-Ang(1-7) = protective.
- Renin ratio >1.5 lateralises; reninoma → HTN + hypokalaemia, partial nephrectomy cures.

I-3d. Patho-physiology of Brain stem death (SAQ)

- $\uparrow$ ICP → Cushing reflex → storm → hypotension, DI, hypothermia, DIC.
- Exclude drugs, hypothermia (<35 °C), metabolic causes.
- Apnoea test: $\text{PaCO}_2 \geq 60$ mmHg, no effort.
- THOA 1994 (2011, Rules 2014) – 4-member board, two tests ≥ 6 h apart; Rule of 100s.

I-3e. Male reproductive immunology (SAQ)

- BTB = Sertoli tight junctions; basal vs adluminal; formed at puberty.
- Privilege also active: M2 macrophages, Tregs, TGF- β , androgens; FasL debated.

- ASA: vasectomy commonest; MAR/IBT $\geq 50\%$ bound = significant; ICSI preferred, steroids out.
- Leukocytospermia >1 million/mL.

I-3f. Principles of antimicrobial therapy in UTI (SAQ)

- Recurrent = $\geq 2/6$ months or $\geq 3/12$ months.
- Cystitis: fosfomycin 3 g single, nitrofurantoin 5 d, pivmecillinam 3–5 d; TMP-SMX 3 d if resistance $<20\%$.
- Pyelonephritis: cipro 7 d / levo 750 mg 5 d (FQ resistance $<10\%$); never nitrofurantoin/fosfomycin.
- ASB: treat only in pregnancy and before mucosa-breaching surgery; aminoglycosides/FQ concentration-dependent, β -lactams time-dependent.